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Lighting Systems (MPA-OSV 25/26L)
Semester Project
Lightning Design of Khamsa Restaurant with Relux Software

Executive Summary

This document outlines the detailed lighting simulation results for the **Khamisa Restaurant** project generated via Relux. The scope covers 9 distinct rooms, optimizing both aesthetic appeal and operational illumination standards.

Overall Project Metrics

- **Number of Rooms:** 9
- **Total Area:** 240.0 m²
- **Total Volume:** 720.0 m³
- **Total Luminaires:** 47 units
- **Total Luminous Flux:** 120,523 lm
- **Total Power Consumption:** 1110.67 W
- **Total Power per Area:** 4.63 W/m²

Detailed Room Analysis

The table below breaks down the spatial dimensions and photometric results for each evaluated area within the restaurant. The metrics include total floor area, power consumption density, maintained illuminance (E_m), and illuminance uniformity (U_o).

Room Description	Area (m ²)	Luminaires	Flux (lm)	Power (W)	E_m (lx)	U_o
Restaurant Hall	120.0	15	44,770	494.0	288	0.35
Bar	12.0	2	8,940	76.0	366	0.73
Manager's Office	12.0	3	7,163	63.7	372	0.40
Restroom W	12.0	6	8,700	66.0	203	0.34
Restroom M	12.0	6	8,700	66.0	214	N/A
Corridor	36.0	4	14,000	116.0	180	0.46
Relaxing Room	12.0	5	7,250	55.0	219	0.91
Storage Room	12.0	2	7,000	58.0	204	0.76
Kitchen	12.0	4	14,000	116.0	541	0.77

Equipment & Parts List

The following luminaire types have been specified for the installation to achieve the performance values stated above:

LTS Cerchio Small

Order No: CERS-P 104.830.1/DALI WH

Specification: 10 W / 594 lm (LED)

ESYLUX ELLEN WCL 300

Order No: EO10850011 (ELLEN WCL 300 OP 1500 830 IP20)

Specification: 11 W / 1450 lm (LED circular area 830)

Glamox A20-S620

Specification: 39 W / 3383 lm (LED 830)

Glamox A35-P555

Order No: A35-P555 4700 830

Specification: 38 W / 4470 lm (LED 830)

Fluora Avola 420

Order No: 13A1.0420.830.6OF (Pendel office)

Specification: 42 W / 5410 lm (LED)

Delta Light BOXY L

Order No: 10726 9228/W (BOXY L 92733 DIM8)

Specification: 6 W / 753 lm (LED ARRAY BR-V8D-G8 2700K)

Delta Light B-LINER

Order No: 388 01 22 (B-LINER 6522 IP)

Specification: 17 W / 2159 lm (LED cluster)

Fluolite FLAT-A 1200

Order No: 88110080 (FLAT-A 1200 OP BASIC 830)

Specification: 29 W / 3500 lm

OPPLE LEDsparrow-E

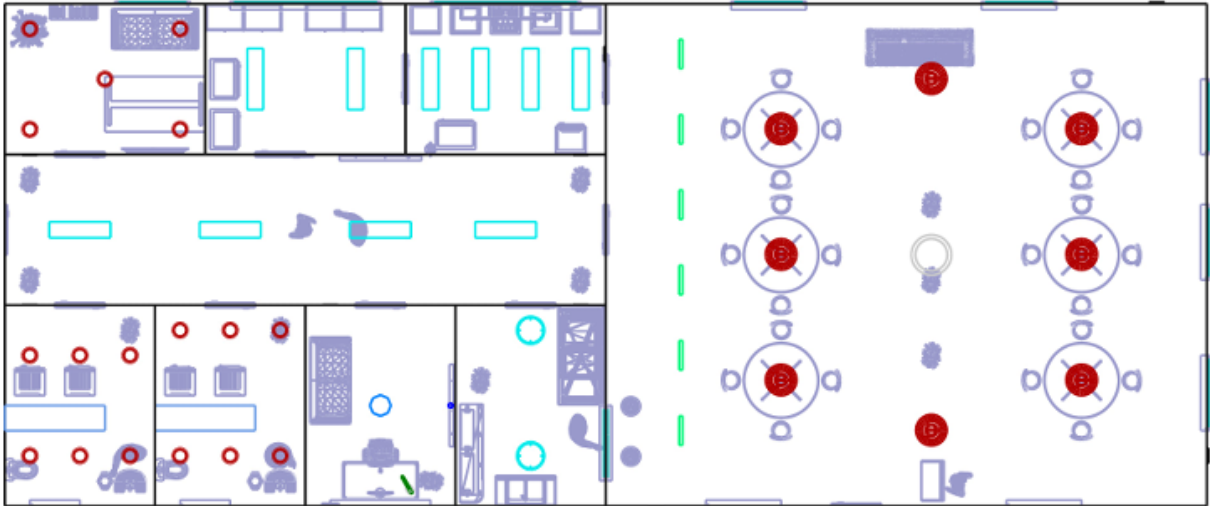
Order No: 523003011600

Specification: 15 W / 1000 lm (DIM-830/840/850)

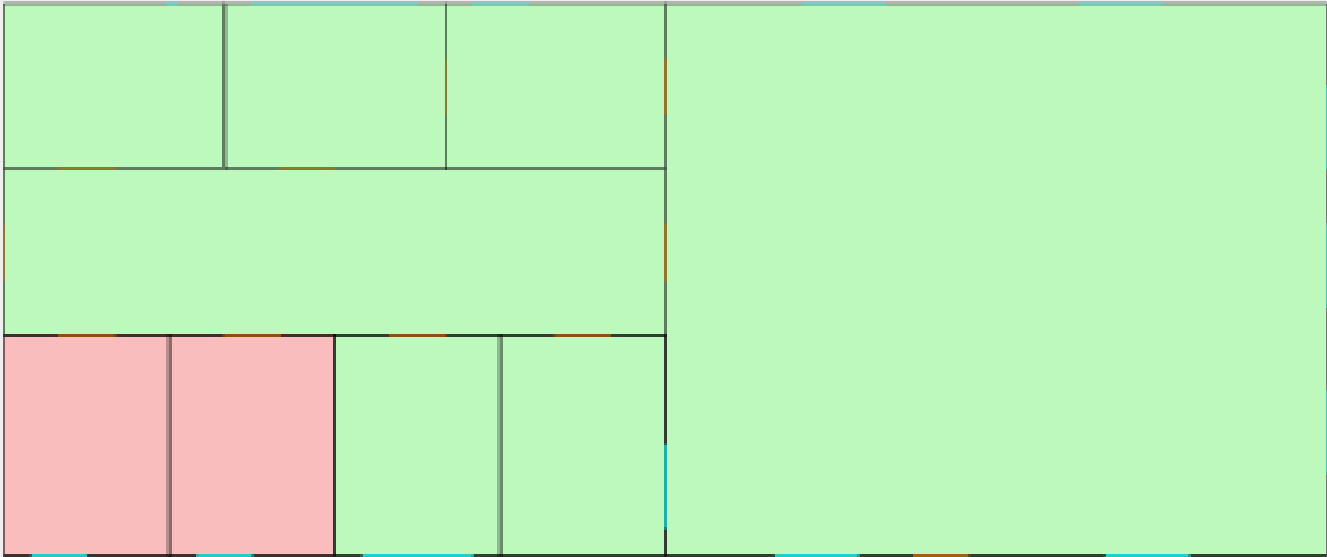
Key Observations

- **Efficiency:** The overall layout maintains a highly efficient power density of 4.63 W/m², showcasing excellent energy management.
- **Task Lighting:** The Kitchen and Offices achieve required task illuminance levels (>500 lx and >300 lx respectively) while minimizing glare.
- **Ambiance:** The Restaurant Hall achieves a comfortable average illuminance of 288 lx, perfectly suited for dining atmosphere requirements.

Floor View:



Floor overview



Number of rooms	9
Total area	240.0 m²
Total volume	720.0 m³
Number of luminaires	47
Total luminous flux	120523 lm
Total power	1110.67 W
Total power per area	4.63 W/m²

- Nominal values met
- Nominal values not met

Parts list

Type No.\Make

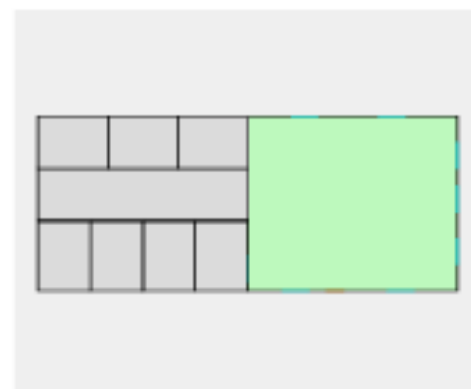
		LTS	
3	1 x	Order No.	: CERS-P 104.830.1/DALI WH
		Luminaire name	: Cerchio Small
		with	: 8 x CERS-P 104.830.1/DALI WH
		Equipment	: 1 x LED 10 W / 594 lm
		ESYLUX	
12	17 x	Order No.	: EO10850011
		Luminaire name	: ELLEN WCL 300 OP 1500 830 IP20
		Equipment	: 1 x LED circular area 830 11 W / 1450 lm

Floor overview

Glamox		
5	8 x	Order No. : A20-S620 LED 3500 830 Luminaire name : A20-S620 Equipment : 1 x LED 830 39 W / 3383 lm
		
6	2 x	Order No. : A35-P555 4700 830 Luminaire name : A35-P Equipment : 1 x LED 830 38 W / 4470 lm
		
Fluora		
10	1 x	Order No. : 13Al.0420.830.6OF Luminaire name : Avola 420 Pendel office Equipment : 1 x LED 42 W / 5410 lm
		
Delta Light		
11	1 x	Order No. : 10726 9228/W Luminaire name : BOXY L 92733 DIM8 Equipment : 1 x LED ARRAY BR-V8D-G8 2700K-cri90-350mA 6 W / 753 lm
		
14	6 x	Order No. : 388 01 22 Luminaire name : B-LINER 6522 IP Equipment : 1 x LED cluster 17 W / 2159 lm
		
Fluolite		
13	10 x	Order No. : 88110080 Luminaire name : FLAT-A 1200 OP BASIC 830 3500lm 29W Equipment : 1 x 1-1172-11p10s/LM281B+/ 1050 29 W / 3500 lm
		
OPPLE		
15	1 x	Order No. : 523003011600 Luminaire name : LEDsparrow-E 15W-DIM-830/840/850 Equipment : 1 x LED 15 W / 1000 lm
		

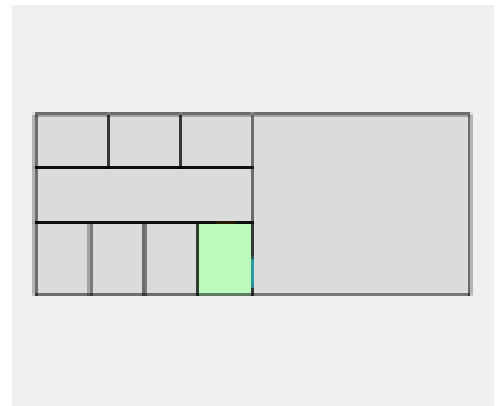
Rooms

Restaurant Hall	15 x Luminaires
Floor area	120.0 m ²
Wall area (without windows)	129.6 m ²
Window area	2.4 m ²
Volume	360.0 m ³
Total luminous flux	44770 lm
Total power	494 W
Total power per area (120 m ²)	4.12 W/m ²
E _m	288 lx
E _{min}	100 lx
E _{min} /E _m (U _o)	0.35
R _{UG}	<=26.3

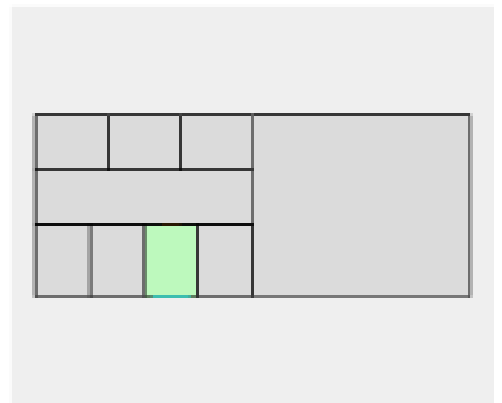


Floor overview

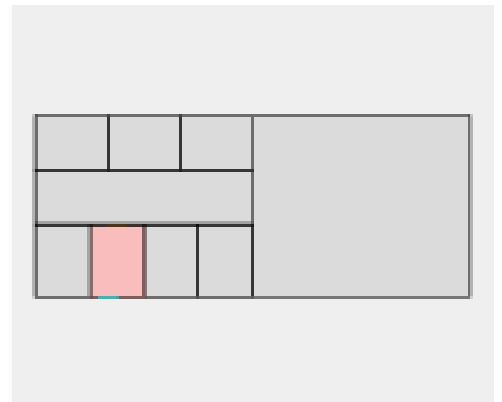
Bar	2 x Luminaires
Floor area	12.0 m ²
Wall area (without windows)	41.7 m ²
Window area	0.3 m ²
Volume	36.0 m ³
Total luminous flux	8940 lm
Total power	76 W
Total power per area (12 m ²)	6.33 W/m ²
$\bar{E}_m$	366 lx (≥ 300 lx)
E_{min}	266 lx
$E_{min}/\bar{E}_m (U_0)$	0.73 (≥ 0.60)
R_{UG}	≤ 18.2 (< 22.00)



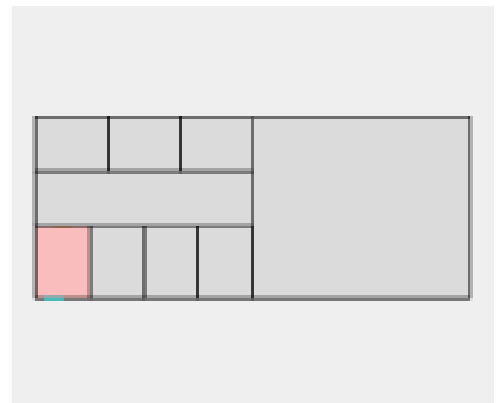
Manager's Office	3 x Luminaires
Floor area	12.0 m ²
Wall area (without windows)	41.7 m ²
Window area	0.3 m ²
Volume	36.0 m ³
Total luminous flux	7163 lm
Total power	63.67 W
Total power per area (12 m ²)	5.31 W/m ²
$\bar{E}_m$	372 lx (≥ 300 lx)
E_{min}	149 lx
$E_{min}/\bar{E}_m (U_0)$	0.40 (≥ 0.40)
R_{UG}	10.0 (< 19.00)



Restroom W	6 x Luminaires
Floor area	12.0 m ²
Wall area (without windows)	41.8 m ²
Window area	0.2 m ²
Volume	36.0 m ³
Total luminous flux	8700 lm
Total power	66 W
Total power per area (12 m ²)	5.50 W/m ²
$\bar{E}_m$	203 lx (≥ 200 lx)
E_{min}	0.0 lx
$E_{min}/\bar{E}_m (U_0)$	— (≥ 0.40)
R_{UG}	≤ 17.4 (< 25.00)



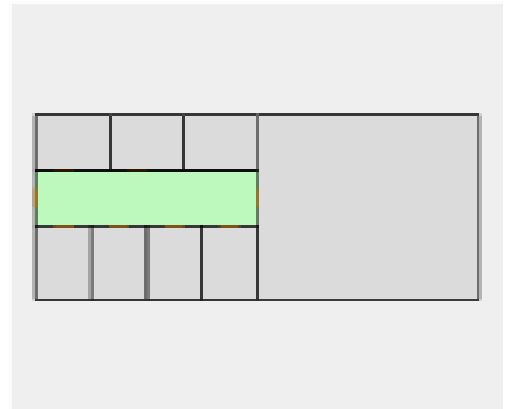
Restroom M	6 x Luminaires
Floor area	12.0 m ²
Wall area (without windows)	41.8 m ²
Window area	0.2 m ²
Volume	36.0 m ³
Total luminous flux	8700 lm
Total power	66 W
Total power per area (12 m ²)	5.50 W/m ²
$\bar{E}_m$	214 lx (≥ 200 lx)
E_{min}	0.0 lx
$E_{min}/\bar{E}_m (U_0)$	— (≥ 0.40)
R_{UG}	≤ 17.4 (< 25.00)



Floor overview

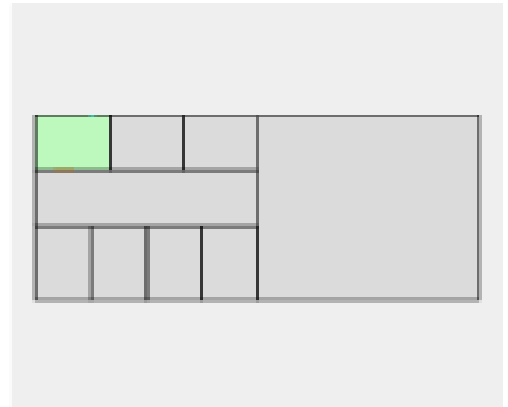
Corridor

Floor area	36.0 m ²	4 x Luminaires
Wall area (without windows)	90.0 m ²	
Window area	0.0 m ²	
Volume	108.0 m ³	
Total luminous flux	14000 lm	
Total power	116 W	
Total power per area (36 m ²)	3.22 W/m ²	
$\bar{E}_m$	180 lx	(≥ 100 lx)
E_{min}	83 lx	
$E_{min}/\bar{E}_m (U_0)$	0.46	(≥ 0.40)
R_{UG}	≤ 20.7	(< 28.00)



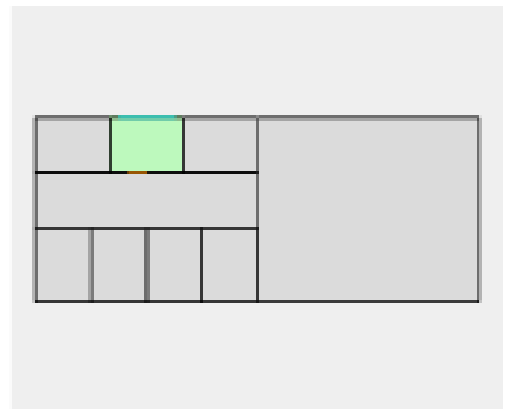
Relaxing Room

Floor area	12.0 m ²	5 x Luminaires
Wall area (without windows)	41.8 m ²	
Window area	0.2 m ²	
Volume	36.0 m ³	
Total luminous flux	7250 lm	
Total power	55 W	
Total power per area (12 m ²)	4.58 W/m ²	
$\bar{E}_m$	219 lx	(≥ 200 lx)
E_{min}	200 lx	
$E_{min}/\bar{E}_m (U_0)$	0.91	(≥ 0.40)
R_{UG}	≤ 17.4	(< 22.00)



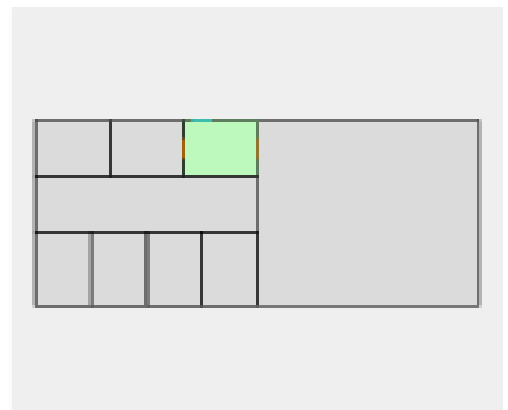
Storage Room

Floor area	12.0 m ²	2 x Luminaires
Wall area (without windows)	41.7 m ²	
Window area	0.3 m ²	
Volume	36.0 m ³	
Total luminous flux	7000 lm	
Total power	58 W	
Total power per area (12 m ²)	4.83 W/m ²	
$\bar{E}_m$	204 lx	(≥ 100 lx)
E_{min}	155 lx	
$E_{min}/\bar{E}_m (U_0)$	0.76	(≥ 0.40)
R_{UG}	≤ 18.0	(< 25.00)



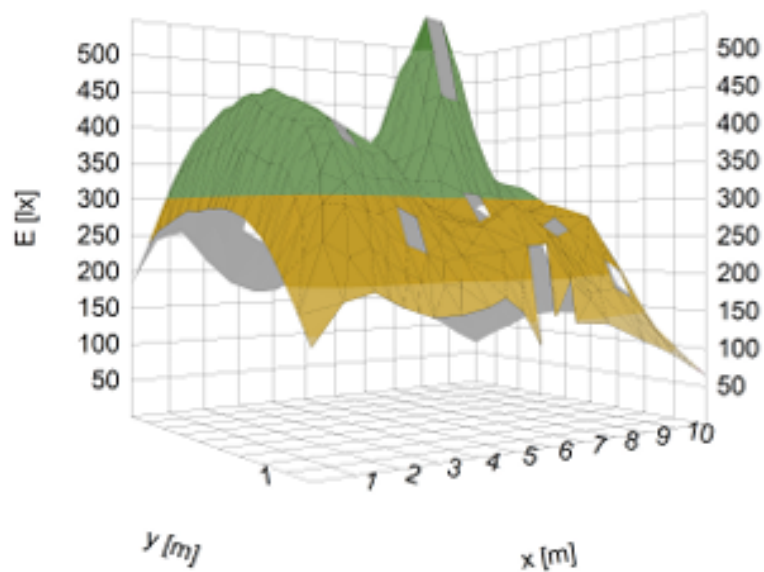
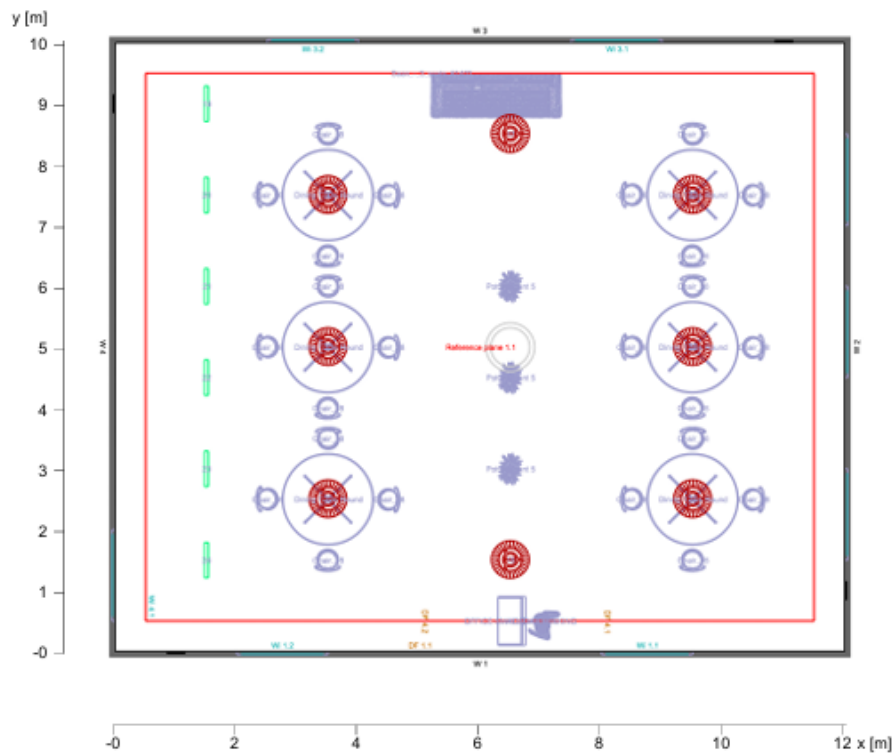
Kitchen

Floor area	12.0 m ²	4 x Luminaires
Wall area (without windows)	41.6 m ²	
Window area	0.4 m ²	
Volume	36.0 m ³	
Total luminous flux	14000 lm	
Total power	116 W	
Total power per area (12 m ²)	9.67 W/m ²	
$\bar{E}_m$	541 lx	(≥ 500 lx)
E_{min}	417 lx	
$E_{min}/\bar{E}_m (U_0)$	0.77	(≥ 0.60)
R_{UG}	≤ 18.0	(< 22.00)



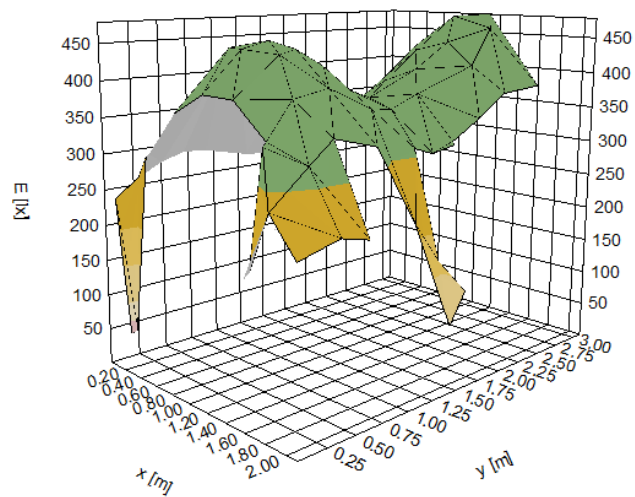
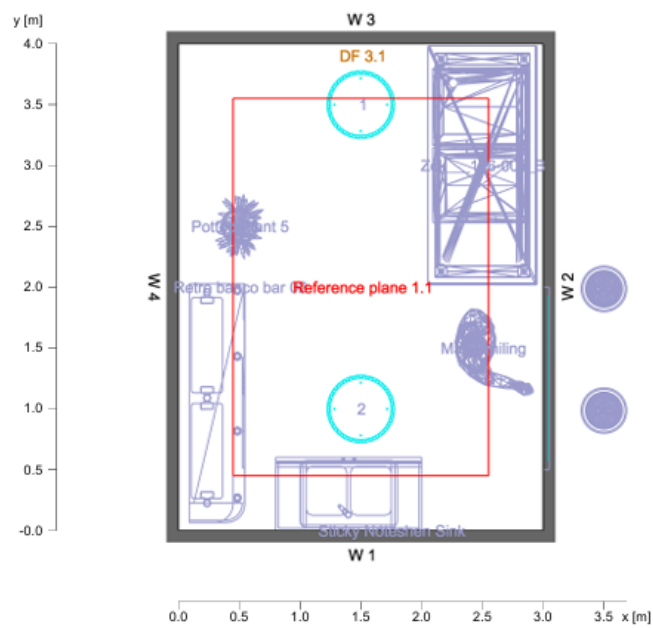
During the calculation process of Restrooms, I faced with some difficulties. As it can be seen from Results Data Sheet. These rooms are meet nominal values, But there is problem with calculating minimum value. That's why it shows that rooms red coloured.

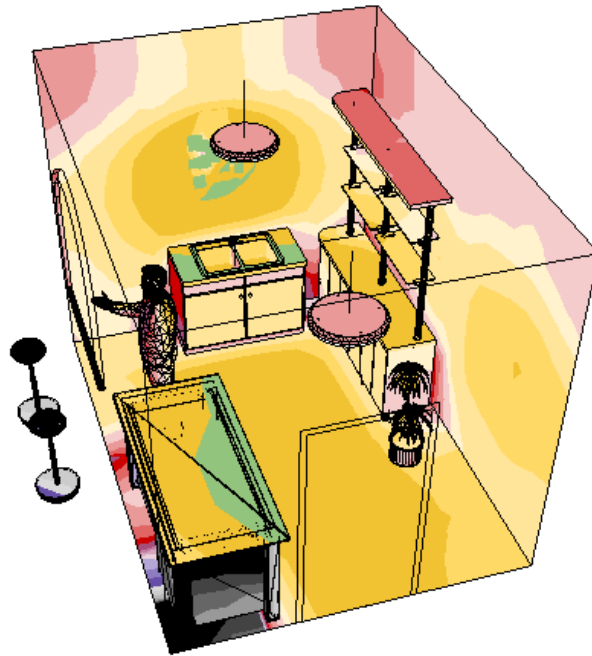
Restaurant



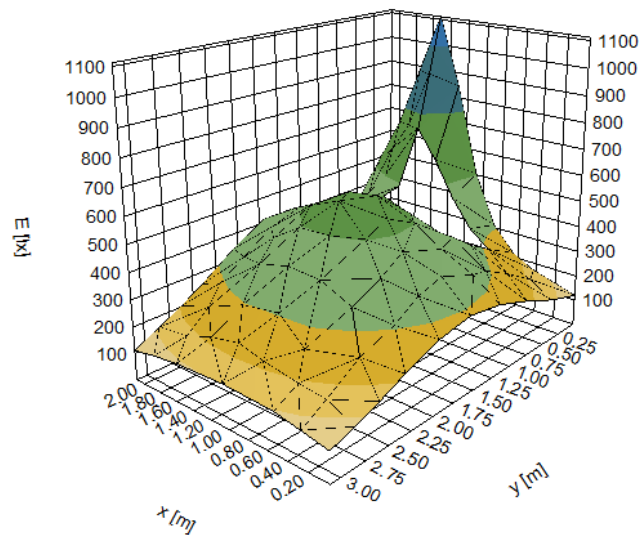
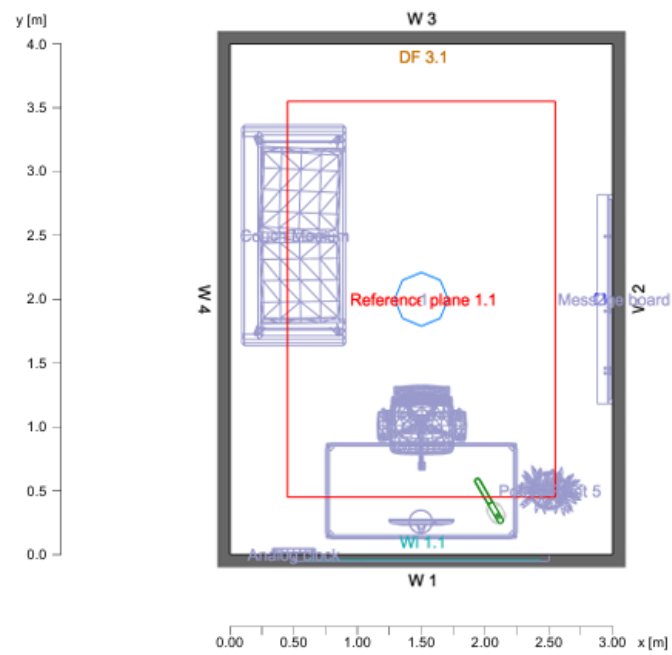


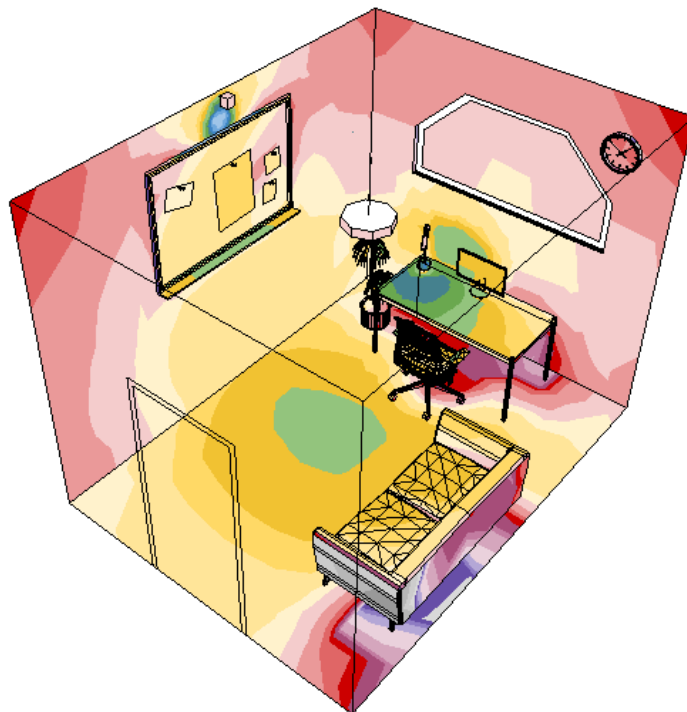
Bar



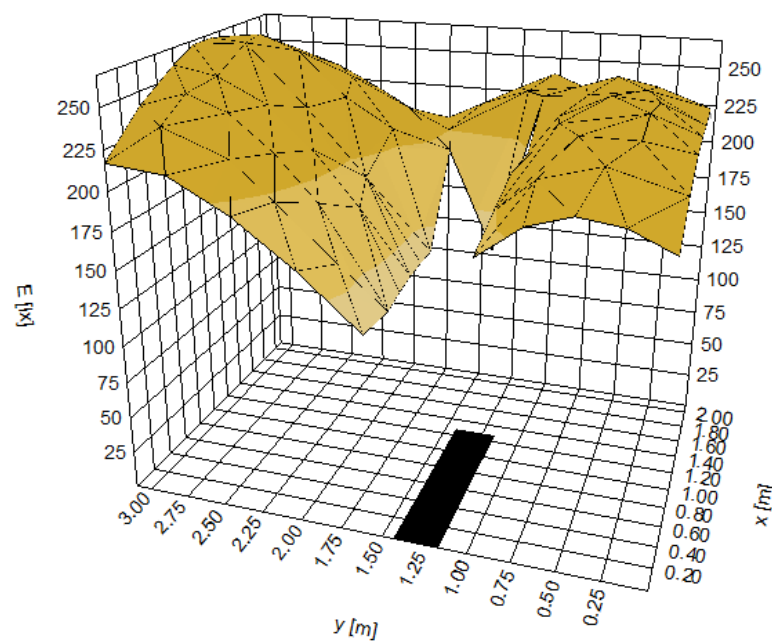
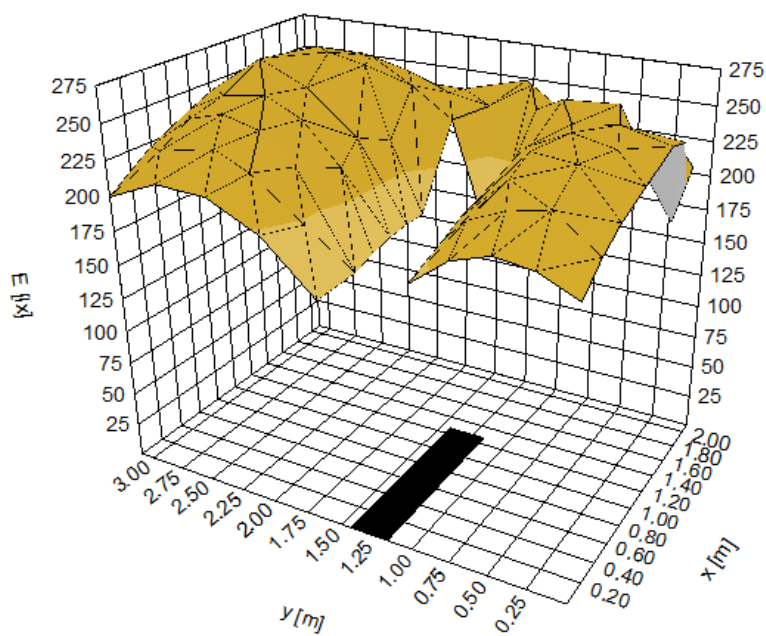
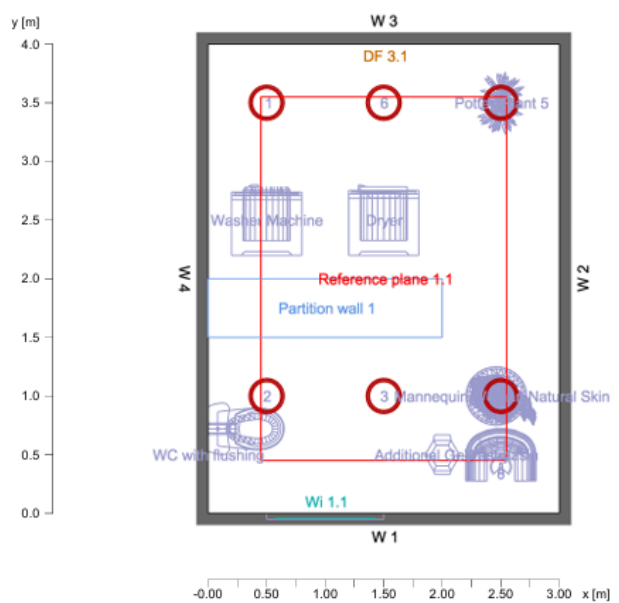
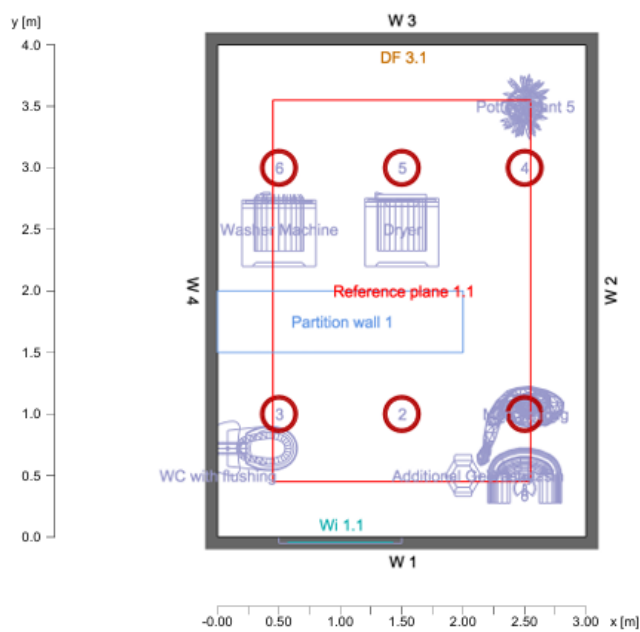


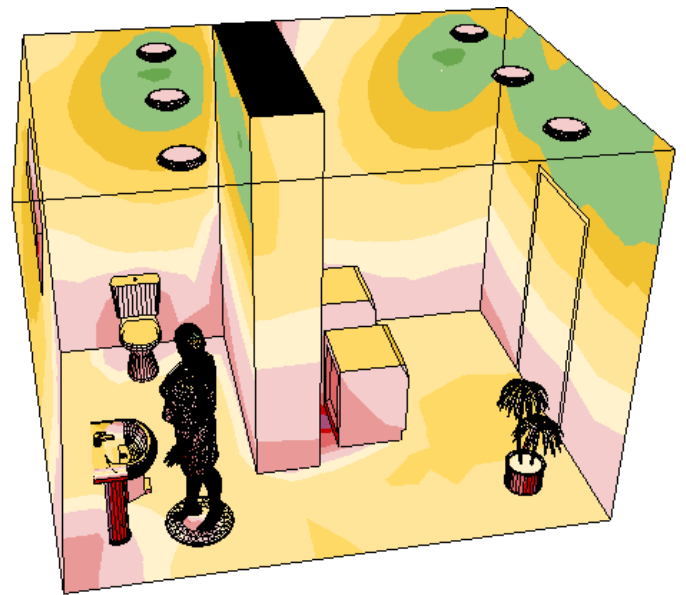
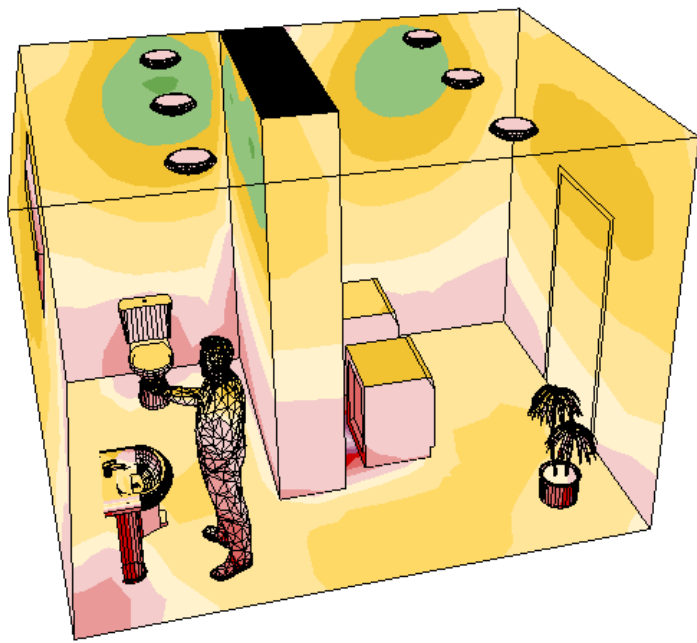
Manager's office



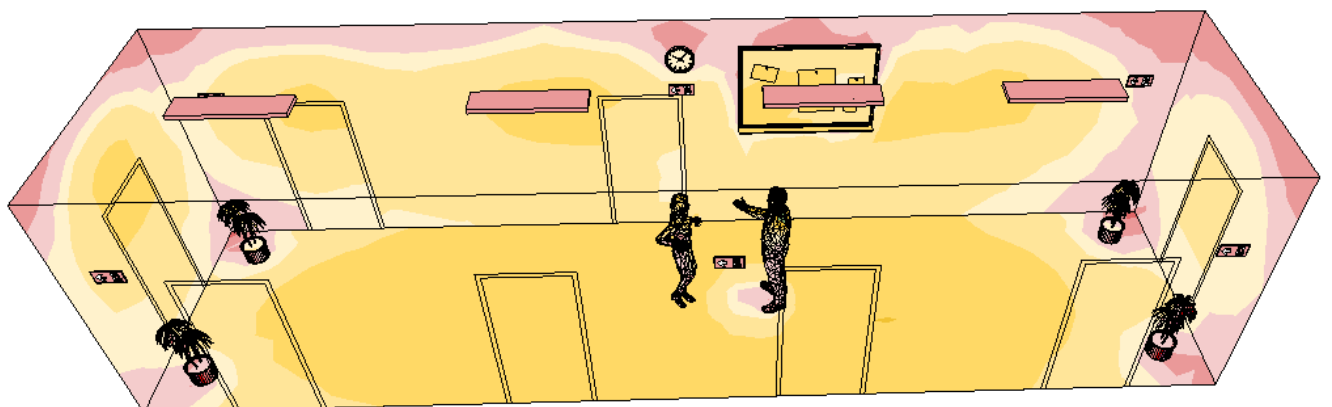
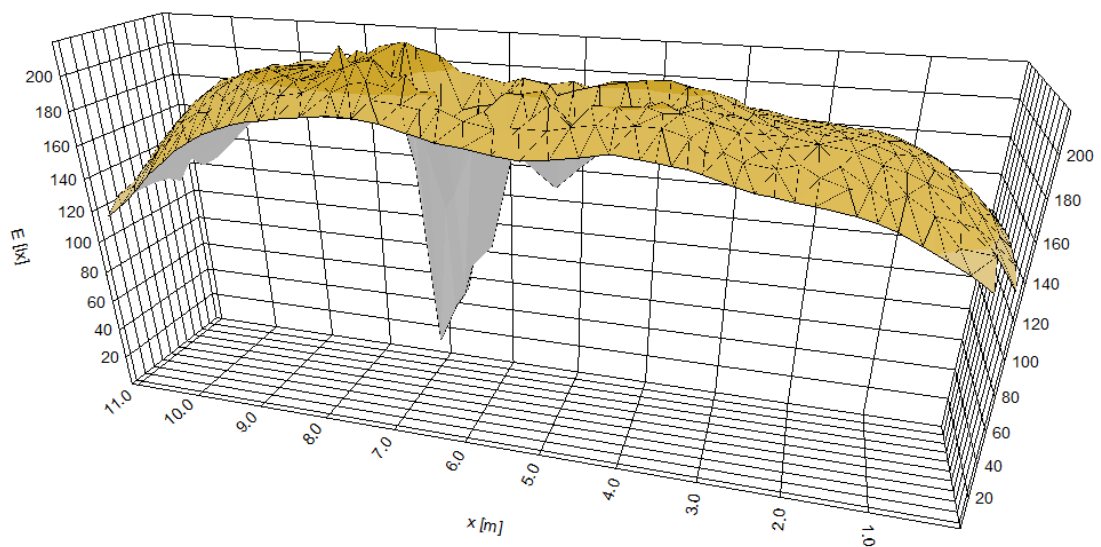


Restrooms

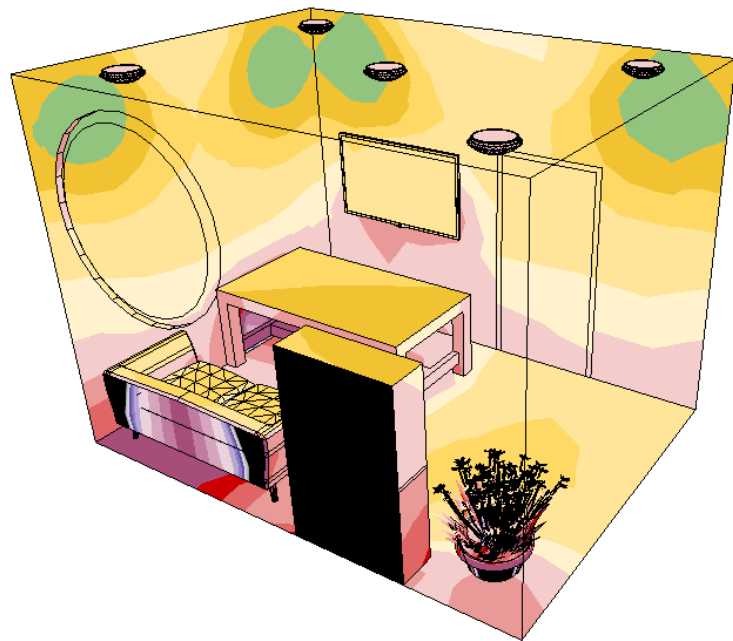
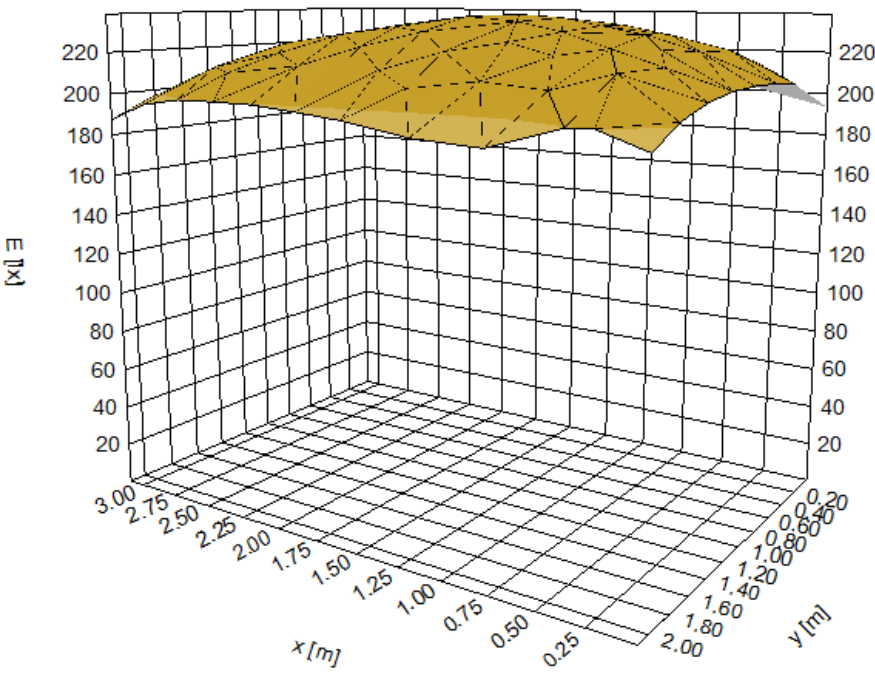
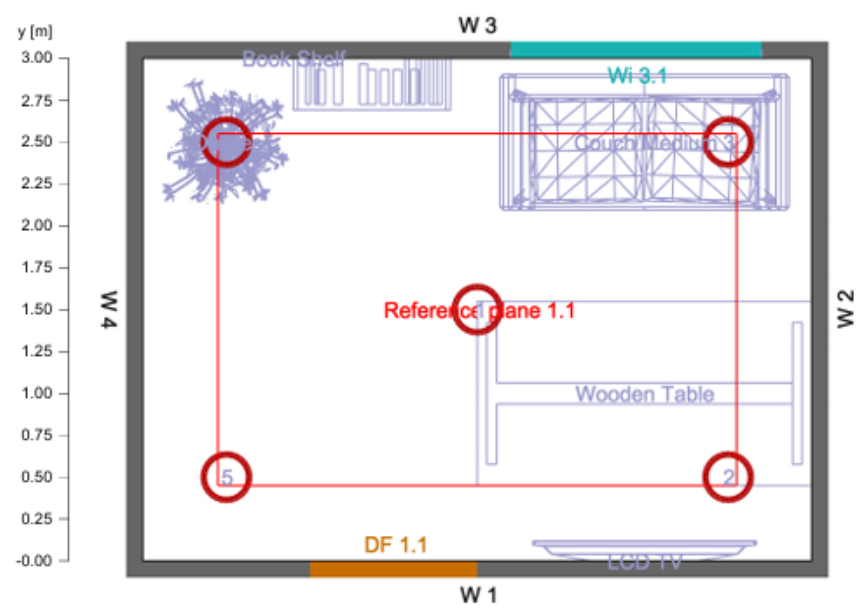




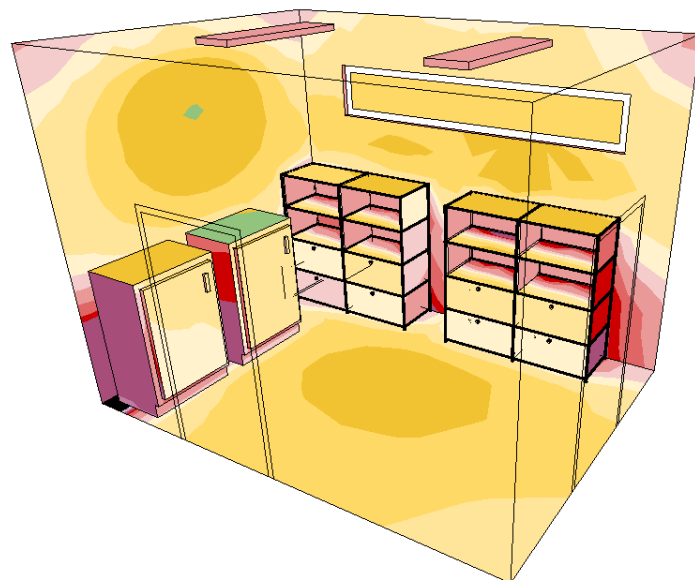
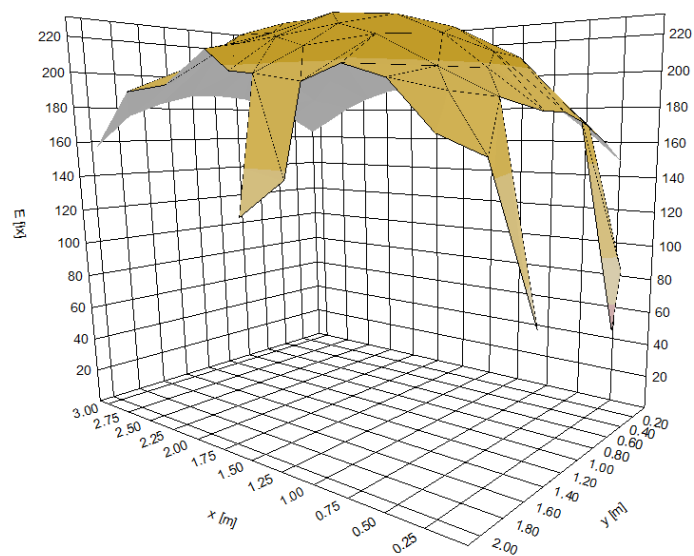
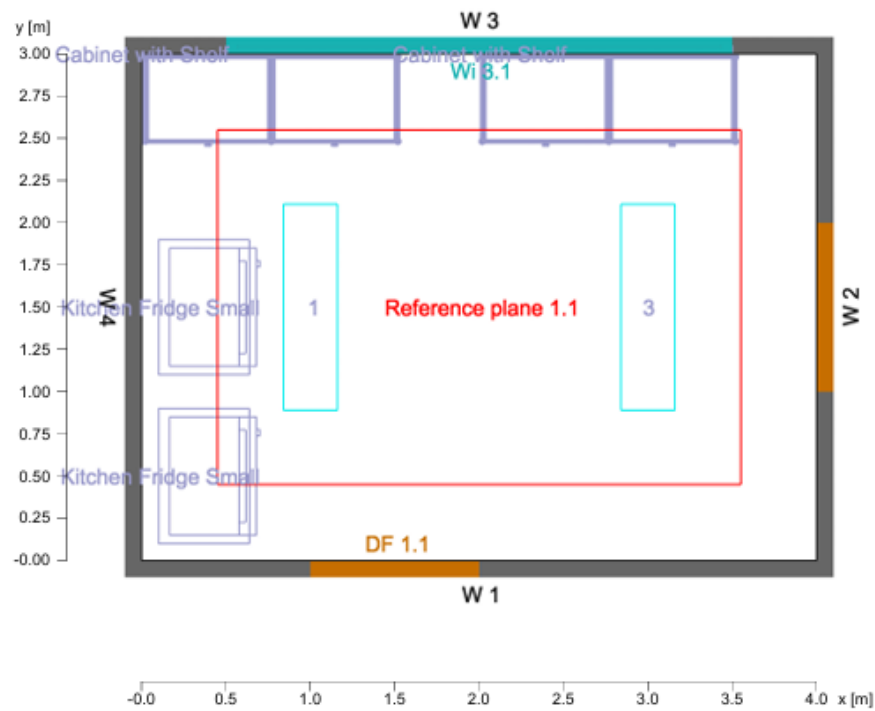
Corridor



Relaxing Room



Storage Room



Kitchen

